Agenda

- Tour of LearnHub
  - https://learn.hub.ku.edu.tr/course/view.php?id=154#section-2
  - Course: INDR252-Fall25-LABA-CE-APPLIED STATISTICS | Learn
- Class exercise
- Boxplots
- Propagation of error

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Class exercise

How can you explain the following statement using the notion of sampling?

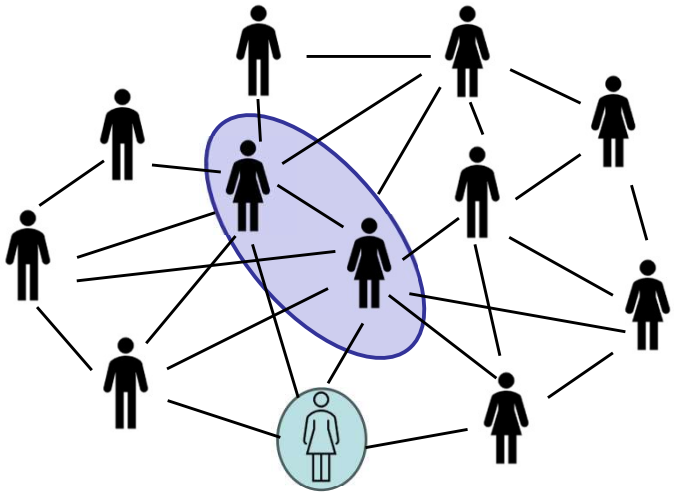
"On average, your friends have more friends than you do."

What kind of assumptions (on the population and on the sample) are needed to justify this statement?

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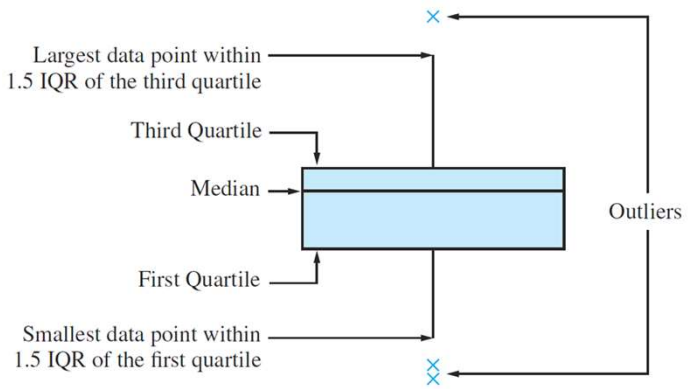
Friendship Paradox

Assumptions:
- The people with many friends constitute a smaller portion of the population.
- I am a randomly chosen person.
- I am more likely to be friends with a person who has many friends.



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Anatomy of a Boxplot

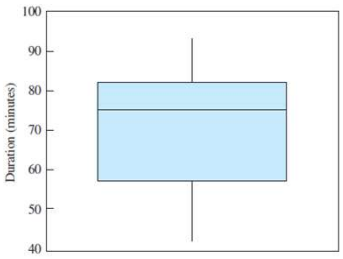


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Example 5

- Notice there are no outliers in these data.
- Looking at the four pieces of the boxplot, we can tell that the sample values are comparatively densely packed between the median and the third quartile.
- The lower whisker is a bit longer than the upper one, indicating that the data has a slightly longer lower tail than an upper tail.
- The distance between the first quartile and the median is greater than the distance between the median and the third quartile.
- This boxplot suggests that the data are skewed to the left.

Stem	Leaf
4	259
5	0111133556678
6	067789
7	01233455556666699
8	000012223344456668
9	013



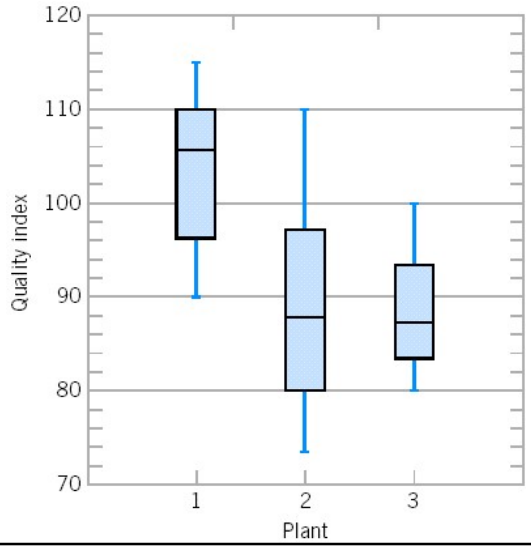
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Comparative Boxplots

- Sometimes we want to compare more than one sample.
- We can place the boxplots of the two (or more) samples side-by-side.
- This will allow us to compare how the medians differ between samples, as well as the first and third quartile.
- It also tells us about the difference in spread between the two samples.

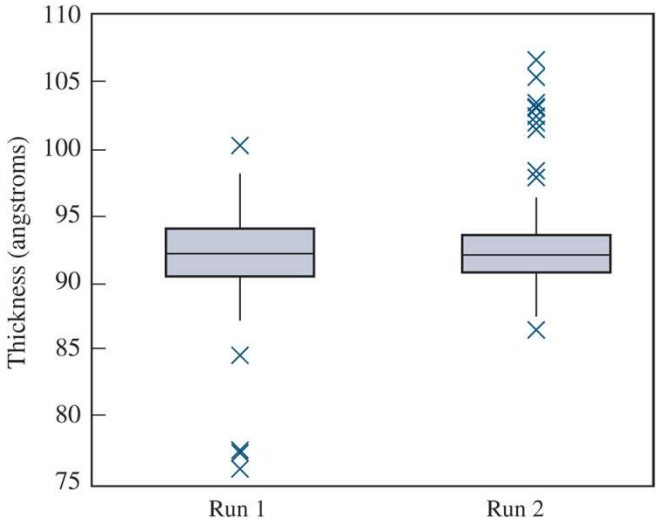
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Comparative box plots of a quality index at three plants



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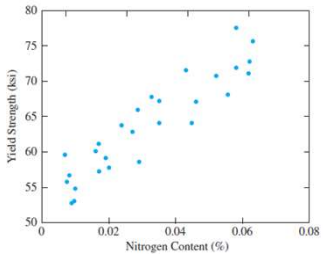
Comparative box plots of oxide layer thickness data



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Scatterplot

- Data for which items consists of a pair of values is called **bivariate**.
- The graphical summary for bivariate data is a **scatterplot**.
- Display of a scatterplot:



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Looking at Scatterplots

- If the dots in the scatterplot do not appear to have a general trend or pattern, then there is little evidence of a relationship between the two variables.
- If the dots on the scatterplot appear to roughly follow a straight line, then there is evidence of a linear relationship between the two variables.
- When a linear relationship exists, we may be able to use one variable to help predict the values of the other variable.

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Summary of Chapter 1

- We discussed types of data.
- We looked at sampling methods, with an emphasis on simple random sampling (SRS).
- We learned about sample statistics.
- We examined graphical displays of data.